

Modular Operator Discovery TSP Report

Overview

- Condition count: 7.
- Best final transfer gap: phase7_full_solver_evolution.
- Best held-out TSPLIB gap: phase7_full_solver_evolution.
- Surviving modular candidates: none.

Run Metadata

- run_name: run_20260520_010243_f
- started_at_local: 2026-05-20 01:02:43
- finished_at_local: 2026-05-20 01:38:21
- duration_hhmm: 00:36
- duration_seconds: 2137.641
- seed_offset: 5000
- replicate_label: f
- judge_status: enabled
- judge_provider: openai
- judge_model: gpt-4.1-mini

Condition Comparison

Condition	Mode	Replay	Selection	Final TSPLIB Gap	Final Family Gap	Final Transfer Gap	Mean Novelty	Mean Complexity	Surviving Candidate	Transplant Delta	Pareto Runtime Inflation
phase7_baseline_heuristic_only	baseline_only	none	score_only	0.235058	0.054968	0.121317	0.0	0.0	False	0.0	0.0
phase7_full_solver_evolution	full_solver	none	score_only	0.083222	0.003941	0.054393	0.840904	0.76	False	0.0	0.0
phase7_modular_operator_evolution	modular_operator	none	score_only	0.235058	0.046321	0.115856	0.826805	0.4425	False	0.005319	1.835159
phase7_modular_operator_random_replay	modular_operator	random	score_only	0.235058	0.046321	0.115856	0.745094	0.445	False	0.00921	0.881022
phase7_modular_operator_diversity_residual_replay	modular_operator	diversity_residual	score_only	0.241011	0.143086	0.179164	0.0	0.445	False	0.027675	24.116978

Condition	Mode	Replay	Selection	Final TSPLIB Gap	Final Family Gap	Final Transfer Gap	Mean Novelty	Mean Complexity	Surviving Candidate	Transplant Delta	Pareto Runtime Inflation
phase7_modular_operator_compression_pressure	modular_operator	none	novelty_gate	0.235058	0.050827	0.118702	0.898566	0.4975	False	0.007362	0.928358
phase7_modular_operator_pareto_selection	modular_operator	none	pareto	0.238386	0.064041	0.128273	0.0	0.4425	False	0.001658	0.014599

Condition Notes

phase7_baseline_heuristic_only

- Execution mode baseline_only on host scaffold nearest_neighbor_2opt.
- Final gaps: TSPLIB 0.235058, family holdout 0.054968, combined 0.121317.
- Accepted novelty 0.0, complexity 0.0, and adaptation efficiency 0.0.
- Last-epoch runtime 0.0 ms and distance evaluations 0.0.
- Validation: surviving False, transplant delta 0.0, positive scaffolds 0, Pareto runtime inflation 0.0.

phase7_full_solver_evolution

- Execution mode full_solver on host scaffold whole_solver.
- Final gaps: TSPLIB 0.083222, family holdout 0.003941, combined 0.054393.
- Accepted novelty 0.840904, complexity 0.76, and adaptation efficiency 0.038881.
- Last-epoch runtime 0.0 ms and distance evaluations 0.0.
- Validation: surviving False, transplant delta 0.0, positive scaffolds 0, Pareto runtime inflation 0.0.

phase7_modular_operator_evolution

- Execution mode modular_operator on host scaffold nearest_neighbor_2opt.
- Final gaps: TSPLIB 0.235058, family holdout 0.046321, combined 0.115856.
- Accepted novelty 0.826805, complexity 0.4425, and adaptation efficiency 0.007.
- Last-epoch runtime 43.07635 ms and distance evaluations 3810.75.
- Validation: surviving False, transplant delta 0.005319, positive scaffolds 2, Pareto runtime inflation 1.835159.
- Validation summary: {"better_gap_same_runtime": false, "code_length_lines": 12, "complexity_score": 0.44, "distance_eval_inflation": 1.785695, "gap_delta": -0.002615, "runtime_inflation": 1.835159, "same_gap_faster": false}

phase7_modular_operator_random_replay

- Execution mode modular_operator on host scaffold nearest_neighbor_2opt.
- Final gaps: TSPLIB 0.235058, family holdout 0.046321, combined 0.115856.
- Accepted novelty 0.745094, complexity 0.445, and adaptation efficiency 0.014726.
- Last-epoch runtime 72.336825 ms and distance evaluations 3810.75.

- Validation: surviving False, transplant delta 0.00921, positive scaffolds 1, Pareto runtime inflation 0.881022.
- Validation summary: {"better_gap_same_runtime": false, "code_length_lines": 12, "complexity_score": 0.44, "distance_eval_inflation": 0.91212, "gap_delta": -0.005462, "runtime_inflation": 0.881022, "same_gap_faster": false}

phase7_modular_operator_diversity_residual_replay

- Execution mode modular_operator on host scaffold nearest_neighbor_2opt.
- Final gaps: TSPLIB 0.241011, family holdout 0.143086, combined 0.179164.
- Accepted novelty 0.0, complexity 0.445, and adaptation efficiency 0.0.
- Last-epoch runtime 44.770675 ms and distance evaluations 3810.75.
- Validation: surviving False, transplant delta 0.027675, positive scaffolds 1, Pareto runtime inflation 24.116978.
- Validation summary: {"better_gap_same_runtime": false, "code_length_lines": 13, "complexity_score": 0.46, "distance_eval_inflation": 628.593581, "gap_delta": 0.057846, "runtime_inflation": 24.116978, "same_gap_faster": false}

phase7_modular_operator_compression_pressure

- Execution mode modular_operator on host scaffold nearest_neighbor_2opt.
- Final gaps: TSPLIB 0.235058, family holdout 0.050827, combined 0.118702.
- Accepted novelty 0.898566, complexity 0.4975, and adaptation efficiency 0.010627.
- Last-epoch runtime 83.88895 ms and distance evaluations 4281.5.
- Validation: surviving False, transplant delta 0.007362, positive scaffolds 1, Pareto runtime inflation 0.928358.
- Validation summary: {"better_gap_same_runtime": false, "code_length_lines": 12, "complexity_score": 0.44, "distance_eval_inflation": 0.911481, "gap_delta": -0.003304, "runtime_inflation": 0.928358, "same_gap_faster": false}

phase7_modular_operator_pareto_selection

- Execution mode modular_operator on host scaffold nearest_neighbor_2opt.
- Final gaps: TSPLIB 0.238386, family holdout 0.064041, combined 0.128273.
- Accepted novelty 0.0, complexity 0.4425, and adaptation efficiency 0.0.
- Last-epoch runtime 101.97515 ms and distance evaluations 3810.75.
- Validation: surviving False, transplant delta 0.001658, positive scaffolds 1, Pareto runtime inflation 0.014599.
- Validation summary: {"better_gap_same_runtime": false, "code_length_lines": 13, "complexity_score": 0.46, "distance_eval_inflation": 0.003718, "gap_delta": 0.006956, "runtime_inflation": 0.014599, "same_gap_faster": false}

Judge Appendix

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TSP Modular-Operator Discovery Suite Review

Overview

- ****Main transfer evidence****: Held-out TSPLIB gap and family holdout gap.
- ****Modular operator interest criterion****: Validation survival after transplant or ablation.

- ****Priority****: Reusable operator structure claims over full-solver performance.
- ****Algorithm discovery****: Only if operator survives scaffold and family tests.

Summary of Conditions

Condition Name	Final Family Gap	Final Transfer Gap	Final TSPLIB Gap	Ablation Support	Transplant Support	Surviving Candidate	Operator Type	Key Notes
phase7_baseline_heuristic_only	0.054968	0.121317	0.235058	N/A	None	No	n/a	Baseline only
phase7_full_solver_evolution	0.003941	0.054393	0.083222	No	No	No	n/a	Full solver; no modular operator
phase7_modular_operator_evolution	0.046321	0.115856	0.235058	Marginal (0.0026)	Weak positive (0.0053)	No	restart_controller	Ablation shows slight gap increase if disabled; partial transplant gains in some scaffolds; high runtime inflation (~1.8x); failed transplant into sparse_three_opt
phase7_modular_operator_random_replay	0.046321	0.115856	0.235058	Marginal (0.0055)	Weak positive (0.0092)	No	restart_controller	Similar to above; slight ablation gap increase if disabled; only 1 positive scaffold; high runtime inflation (0.88x)
phase7_modular_operator_diversity_residual_replay	0.143086	0.179164	0.241011	Negative effect (disabling improves gap by ~0.0578)	Negative (gap worsens with operator transplant, +0.0277)	No	scaffold_selector	Operator consistently underperforms baseline (esp. heldout TSPLIB and many families); ablation does not support effect; numerous transplant failures; very high runtime inflation (24x)
phase7_modular_operator_compression_pressure	0.050827	0.118702	0.235058	Marginal (0.0033)	Weak positive (0.0074)	No	restart_controller	Similar pattern to restart controllers above; no significant ablation impact; failed transplant to sparse_three_opt; runtime inflation ~0.93

Condition Name	Final Family Gap	Final Transfer Gap	Final TSPLIB Gap	Ablation Support	Transplant Support	Surviving Candidate	Operator Type	Key Notes
phase7_modular_operator_pareto_selection	0.064041	0.128273	0.238386	No (disabling improved gap by 0.0069)	Negative (gap delta +0.0017)	No	candidate_ranker	No clear validation; underperformed in grid-like family; multiple transplant failures; minimal positive signals

Conservative Interpretation

1. ****Full Solver and Baseline Conditions****

- Full solver evolution condition shows good family and transfer gaps (0.0039 fam gap, 0.054 transfer gap), but no modular operator component.
- Baseline is significantly worse on all fronts.

2. ****Modular Operators of type "restart_controller"****

- Appear in multiple related conditions (modular_operator_evolution, random_replay, compression_pressure).
- Ablation results:
 - Slight (but small) performance degradation when disabled, supporting some operator effect.
 - Variants with fixed or wider restart budgets show no gap changes vs. full.
- Transfer:
 - Moderate to weak positive transplant mean gap deltas (~0.005-0.009), only 1-2 positive scaffolds out of 5.
- Family holdout gap improvements are modest and concentrated in "clustered_tsp" and "two_cluster_bottleneck_tsp" families.
- Failure to transplant cleanly into some scaffolds (especially sparse_three_opt) limits reuse claim.
- Runtime inflation notable (around 0.88 to 1.83 times), reducing practicality.
- Final TSPLIB gaps remain high (~0.235), same as baseline for heldout TSPLIB instances.

****Conclusion:****

The restart_controller modular operators show some limited reusable structure signal for bottleneck and clustered TSP families but fail strong transplant and ablation checks to claim discovery or robust reuse. They are not surviving candidates under strict criteria.

3. ****Scaffold Selector Operator****

- Shows consistent underperformance vs. baseline on held-out TSPLIB and most families.
- Ablation indicates disabling improves performance notably (gap delta -0.058).
- Transplant attempts failed in multiple scaffolds.

- Extremely high runtime inflation (~24x) further disqualifies.
- No survival evidence to claim operator efficacy.

****Conclusion:****

Not interesting under all criteria; no validation support.

4. ****Candidate Ranker Operator****

- No consistent family or transfer improvement.
- Ablation shows disabling operator improves performance.
- Transplant results are slightly negative.
- Underperforms on grid-like family and no families show clear gain.
- Not validated under transplant or ablation.

****Conclusion:****

No reusable or transferable operator discovered here.

Final Evaluation and Recommendations

Criterion	Result Summary	Interpretation
Held-out TSPLIB Gap	Modular ops show no improvement over baseline	No transfer improvement on hard set
Family Holdout Gap	Restart controllers show small gains in 2 families	Limited, non-robust effect
Transplant Results	Weak positive gap deltas in restart controllers	Partial transfer; not consistent
Ablation Tests	Marginal or negative impact on operator removal	Weak modular operator effect
Pareto Tradeoffs	Runtime inflation high; Pareto improvements absent	Unfavorable cost-benefit balance
Survival as Candidate	None survived; all flagged as non-surviving	No confirmed reusable modular ops

****Summary:****

- No modular operator satisfies transplant and ablation validation criteria robustly enough to claim discovery or reusable operator structure.
- Restart controllers provide weak evidence of reusable restart pressure control aiding bottlenecked TSP instances, but fail transplant into key scaffolds and exhibit high runtime cost.
- Other modular types (scaffold selector, candidate ranker) demonstrate negative or no transfer signals, ablation failures, and worse runtime profiles.
- Full-solver evolution yields best overall gaps but lacks modular operator confirmation.
- Conservative interpretation: No modular operator discovered that reliably transfers across families and scaffolds with validated improvement.

- Recommend revisiting modular designs focused on restart strategies with improved runtime efficiency and stronger transplant validation.
